

Ahmed Mohamed

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Education

Bachelor of Science, Information Technology AI Driven Solutions + Cybersecurity

Graduation Year: 2027

George Mason University – Fairfax, VA.

Skills

Programming, Full-Stack Development, & APIs: Python, TypeScript, JavaScript, SQL, Java, Node.js, React, Next.js, Electron, REST API Design, API Integration, JWT/OAuth Authentication

AI & Machine Learning: Scikit-Learn, Retrieval-Augmented Generation (RAG), Agentic AI Orchestration, LLM Integration, Feature Engineering, Model Evaluation, ML Pipelines

Databases & Infrastructure: PostgreSQL, SQLite, AWS EC2, Azure

Cybersecurity & Data Protection: Network Analysis (Wireshark), Nmap, Shodan, Secure Data/PII Handling, Threat Detection

Experience

CodePath AI Open Source Capstone Fellow

Open Source Engineering, AI Development Workflows, Git/GitHub, Pull Requests, Code Review, LLM-Assisted Development

Jun 2026

- Contributing to a real-world open source AI software project through CodePath's AI engineering practicum, working in an unfamiliar production codebase with Git/GitHub, issue scoping, local setup, debugging, testing, and pull request workflows.
- Applying AI-assisted engineering practices for repository analysis, code comprehension, scaffolding, and debugging while maintaining full ownership of implementation quality, documentation, and maintainer review readiness.

DoD National Defense Technologies Hackathon (NDIA) - Top 5 Finalist

Python, Scikit-Learn, Flask, JavaScript (Chrome Extension APIs), REST APIs, AWS EC2

Aug 2025

- Developed and deployed a machine learning phishing detection Chrome extension as a Top 5 Finalist at the DoD National Defense Technologies Hackathon, achieving **90%** accuracy on **1.1M+** publicly sourced URL and metadata data points using Python, Scikit-Learn, Flask, JavaScript, REST APIs, and AWS EC2.
- Engineered a real-time detection pipeline for analyzing URLs and encoded payloads in a browser-based environment, applying machine learning to automated threat detection, large-scale risk identification, and Department of Defense cybersecurity use cases.

CodePath Systems & Security Technical Fellow

Python, Linux, Network Analysis, Wireshark, Nmap, Shodan, SET Toolkit

May 2025 – Aug 2025

- Completed **15+** hands-on cybersecurity labs focused on network traffic analysis, intrusion detection, and vulnerability identification using tools such as Wireshark, Nmap, and SIEM-based monitoring systems.
- Simulated real-world attack and defense scenarios, including phishing analysis, packet inspection, and incident response workflows, developing practical experience in threat detection and system hardening.

Projects

Founder & AI Product Engineer - Skillverse (Adaptive AI/ML Learning Platform)

Next.js, TypeScript, PostgreSQL, LLM APIs, Retrieval-Augmented Generation (RAG), Multi-Agent Architecture, JWT/OAuth

Feb 2026 - Ongoing

- Developing an AI-powered learning platform with **1000+** waitlist users that generates personalized courses, quizzes, flashcards, and pace-adaptive study tracks using a multi-agent RAG pipeline, self-evaluating curriculum generation, and adaptive learner profiles to personalize content, pacing, and difficulty.

Owner & AI Automation Manager - CarbonML (AI Voice & Business Operations Service)

Python, OpenClaw, Retell AI, n8n, Twilio, ElevenLabs, Stripe, Google APIs, AI Agents, Workflow Automation

Dec 2025 - Ongoing

- Operate CarbonML as a AI voice automation business that helps companies handle inbound/outbound calls, evaluate caller intent, qualify leads, answer customer questions, schedule appointments, support multilingual conversations, process handoffs, and escalate complex cases to a human agent. Using OpenClaw in a supervisor-agent architecture to automate owner-level operations such as lead generation, outreach/marketing, website development/updates, email support, billing workflows, compliance checks, and operational monitoring.

Founder & Full-Stack Product Engineer - DueD8 (Academic Intelligence Platform)

React, Node.js, TypeScript, PostgreSQL, REST APIs, Canvas LMS API, OpenAI API, Secure Authentication(JWT/OAuth)

Oct 2025 - Ongoing

- Built and deployed an AI-powered academic platform that automates syllabus, assignment, and grade parsing through Canvas LMS API integrations, generating personalized study guidance with **95%** parsing accuracy and improving study efficiency by **75%** through secure tokenized data pipelines.

Automation Developer - EIP Assist (Secure University Communications Automation System)

Python, n8n, Microsoft Graph API, REST APIs, Secure Automation Pipeline, FERPA-compliance

Sep 2025

- Engineered a secure FERPA-compliant university communications automation system integrating Python, n8n, and Microsoft Graph API to generate GMU policy-aligned responses using staff-updated program data; reducing manual email workload by **85%+** and improving student/family response times by **90%**.